

# Efficacy and mechanism of actions of cipargamin as an antibabesial drug candidate

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## Abstract

Babesiosis is a disease brought on by intraerythrocytic parasites of the genus *Babesia*. Current chemotherapies are accompanied by side effects and parasite relapse. Therefore, it is crucial to develop highly effective drugs against *Babesia*. Cipargamin (CIP) has shown inhibition against apicomplexan parasites, mainly *Plasmodium* and *Toxoplasma*. This study evaluated the growth-inhibiting properties of CIP against *Babesia* spp. and investigated the mechanism of CIP on *B. gibsoni*. The half inhibitory concentration (IC<sub>50</sub>) values of CIP against the *in vitro* growth of *B. bovis* and *B. gibsoni* were 20.2 ± 1.4 nM and 69.4 ± 2.2 nM, respectively. CIP significantly inhibited the growth of *B. microti* and *B. rodhaini* *in vivo*. Resistance was conferred by L921V and L921I mutations in *BgATP4*, which reduced the sensitivity to CIP by 6.1- and 12.8-fold. An *in silico* investigation revealed reductions in affinity for CIP binding to *BgATP4*<sup>L921V</sup> and *BgATP4*<sup>L921I</sup> compared to *BgATP4*<sup>WT</sup>. In this study, we characterized the efficacy of CIP against *Babesia* spp. by investigating the mechanistic basis for the resistance to CIP conferred by mutations in the *BgATP4*. Our findings present a promising starting point for the establishment of new therapeutic interventions against *Babesia* infection.

### eLife assessment

This study presents **valuable** findings with practical and theoretical implications for drug discovery, particularly in the context of repurposing CIP for the treatment of *Babesia* spp. The evidence is **convincing** overall, as the data and analyses support the main claims. However, a few assertions are only partially substantiated. If the authors can strengthen these areas with additional evidence, the paper could attract greater interest from scientists in drug discovery, computational biology, and microbiology.

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## Introduction

*Babesia* is an apicomplexan tick-transmitted hemoparasite, which not only impacts the livestock economy but also causes an emerging disease in humans (Jalovecka et al., 2019 [↗](#)). *Babesia* is a global pathogen that is more prevalent in certain regions such as Asia, Europe, and North America. However, as climate change brings with it higher temperatures and humidity, ticks and their reservoir hosts are anticipated to expand northward for survival and activity (Drews et al., 2023 [↗](#)). The disease it causes is known as babesiosis, commonly characterized by fever and hemolytic anemia, but chronic infections are asymptomatic (Almazán et al., 2022 [↗](#)). In acute cases, up to 20% of patients die, particularly individuals with compromised immune systems (Krause, 2019 [↗](#)).

Currently, the most common drugs for the treatment of human babesiosis include a combination of atovaquone (ATQ) plus azithromycin (AZM) or clindamycin plus quinine (Holbrook et al., 2023 [↗](#)). Still, these recommended treatments often result in various problems. For instance, multiple mutations in the *B. microti* cytochrome b, which was associated with the atovaquone-target region, were discovered in an atovaquone-treated relapse patient (Holbrook et al., 2023 [↗](#)). The combination of clindamycin and quinine is the last resort for patients with severe symptoms. Despite its efficacy, some had adverse drug reactions after taking these drugs (Vannier and Krause, 2012 [↗](#)). The 8-aminoquinoline analog, tafenoquine (TAF), was found to be effective in curing the patient's *B. microti* infection that was resistant to azithromycin and atovaquone with a 200 mg weekly dose (Marcos et al., 2022 [↗](#)). However, the efficacy of TAF treatment may vary between individuals and cases and its contraindication in G6PD-deficient patients limits its clinical use. Due to the aforementioned issues, it is undoubtedly urgent to search for compounds that treat infections caused by *Babesia* spp. while simultaneously minimizing the detrimental effects of antibabesial drugs.

Spiroindolone cipargamin (CIP), a promising antimalarial, has been found to effectively suppress the growth of all strains of *Plasmodium falciparum* and *P. vivax* with potency in the low nanomolar ranges (Rosling et al., 2018 [↗](#); Rottmann et al., 2010 [↗](#)). The assessment of the pharmacokinetic characteristics of the drug indicated that oral administration of CIP had good absorption, a prolonged half-life, and exceptional bioavailability (Schmitt et al., 2022 [↗](#)). Following oral CIP treatment (30 mg daily for three days) in adults with simple *P. falciparum* or *P. vivax* malaria, the parasitemia was rapidly cleared in a phase II trial (Schmitt et al., 2022 [↗](#)). Currently, a clinical trial is being conducted for the intravenous administration of CIP as a treatment for severe malaria patients (ClinicalTrials, 2024 [↗](#)). CIP was also evaluated for treating toxoplasmosis as a second-line medicine in cases where there are intolerable toxicity issues or allergies to the currently used treatments. Five days post-*Toxoplasma gondii* infection, CIP-treated mice (100 mg/kg/day on the day of and the day after infection) had 90% fewer parasites than the control mice (Zhou et al., 2014 [↗](#)).

Due to its excellent efficacy against other apicomplexan parasites, we hypothesize that CIP may also be effective as an antibabesial drug. Therefore, the objectives of this study were to determine whether CIP could inhibit the growth of *Babesia* spp., namely *B. bovis* and *B. gibsoni* *in vitro* and *B. microti* and *B. rodhaini* *in vivo*, and identify the inhibitory mechanisms of CIP on *Babesia* parasites (*BgATP4*) by monitoring the Na<sup>+</sup> and H<sup>+</sup> balance.

## Results

### Inhibitory efficacy of CIP on *B. bovis* and *B. gibsoni* *in vitro*

The *in vitro* activity of CIP against *B. bovis* and *B. gibsoni* showed a steep growth inhibition curve with half inhibitory concentration ( $IC_{50}$ ) values of  $20.2 \pm 1.4$  nM (Figure 1A) and  $69.4 \pm 2.2$  nM (Figure 1B), respectively. The  $CC_{50}$  value of CIP on MDCK cells was  $38.7 \pm 2.0$   $\mu$ M (Figure supplement 1), resulting in a predicted selectivity index of 557.6. Furthermore, at a concentration of 100  $\mu$ M, CIP exhibited a low erythrocyte hemolysis rate of  $0.11 \pm 0.03$  % (not shown).

### CIP effect on *B. microti* and *B. rodhaini* infections *in vivo*

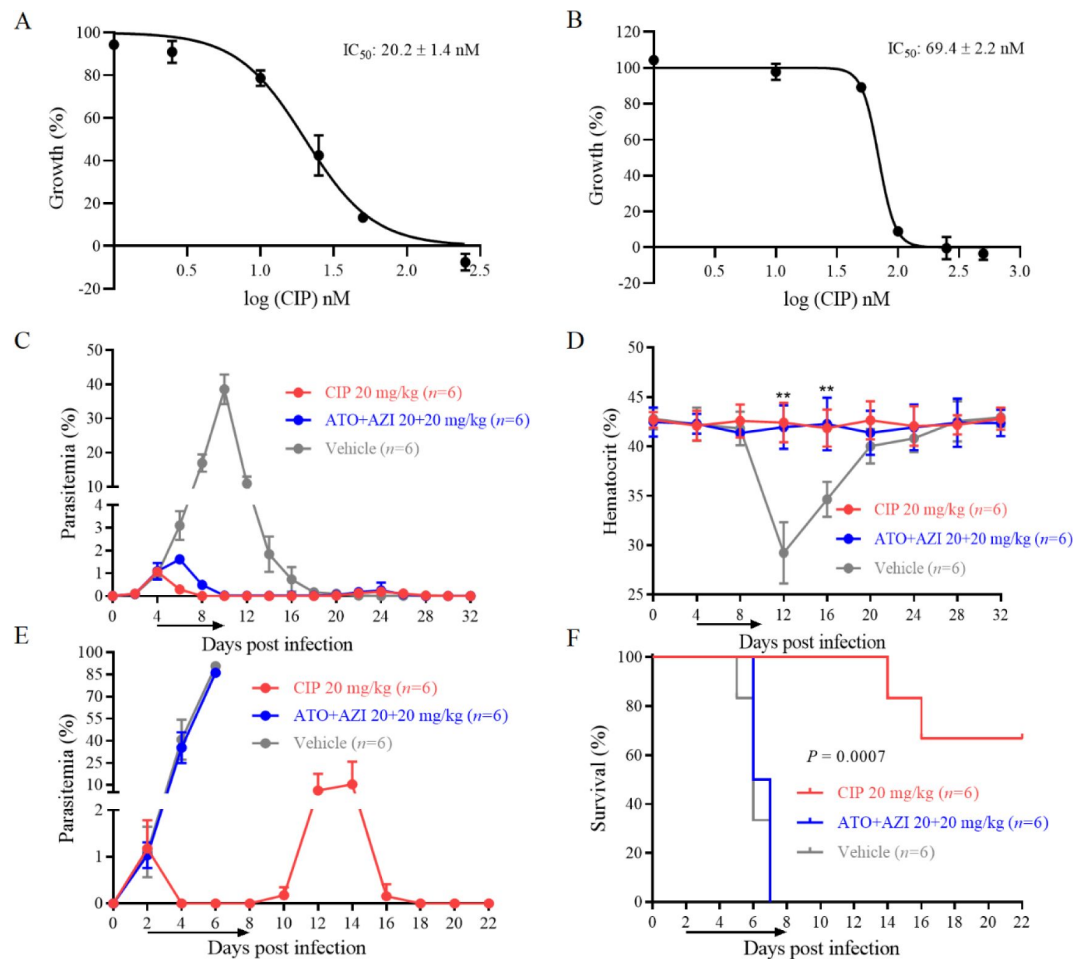
Concurrently, CIP showed effective inhibition on *B. microti* and *B. rodhaini* *in vivo*. The parasitemia of *B. microti*-infected BALB/c mice grew dramatically in the vehicle-treated control group and peaked at 10 DPI ( $35.88 \pm 4.32\%$ ) (Figure 1C). On the other hand, the treatment with CIP (20 mg/kg of body weight) or atovaquone (ATO) plus azithromycin (AZI) administered orally resulted in a significantly lower peak parasitemia,  $1.06 \pm 0.20\%$  and  $1.61 \pm 0.20\%$ , respectively. Hematocrit (HCT) variations were tracked every 4 days as an indicator of anemia in *B. microti*-infected mice. The vehicle-treated group showed a drop in HCT levels from 8 DPI to 24 DPI, with statistically significant decreases at 12 and 16 DPI ( $P < 0.01$ ). No significant reduction in HCT levels was observed in the CIP-treated group or the ATO plus AZI-treated group (Figure 1D). This suggests that the administration of CIP could control *B. microti* infection and prevent anemia from developing in *B. microti*-infected mice. BALB/c mice infected with *B. rodhaini* treated with sesame oil or ATO plus AZI showed high parasitemia,  $90.73 \pm 1.97\%$ , and  $86.23 \pm 3.06\%$ , respectively (Figure 1E), and all mice died within 10 DPI (Figure 1F). CIP treatment in *B. rodhaini*-infected mice precluded the emergence of parasitemia for the following eight days (Figure 1E), which led to 66.67% of mice surviving the challenge infection (Figure 1F). At 14 DPI, parasites had recurred in all CIP-treated *B. rodhaini*-infected mice ( $10.32 \pm 15.51\%$ ), which were eventually cleared as indicated by undetectable parasites at 18 DPI (Figure 1E).

### Identification of *B. gibsoni* ATP4 mutation in CIP-resistant strains

We sequenced the *B. gibsoni* ATP4 gene from two resistant parasite wells. A single nucleotide variant (SNV) in *BgATP4* with a substitution at position 2,761 (from C to G) was found – a nonsynonymous coding change from leucine to valine (L921V) (Figure 2B). In another resistant strain, the mutation showed in the same position. However, the nucleotide substitution was from C to A, and the coding changed from leucine to isoleucine (L921I) (Figure 2C). *BgATP4*<sup>L921V</sup> and *BgATP4*<sup>L921I</sup> lines were tested for their susceptibility to CIP, and had  $IC_{50}$  values of  $421.0 \pm 15.88$  nM and  $887.9 \pm 61.97$  nM, respectively (Figure 2D). These findings demonstrate a 6.1- and 12.8-fold reduction in CIP sensitivity of resistant parasite lines *BgATP4*<sup>L921V</sup> and *BgATP4*<sup>L921I</sup>.

### The effect of CIP on *BgATP4*<sup>WT</sup>, *BgATP4*<sup>L921V</sup> and *BgATP4*<sup>L921I</sup> function

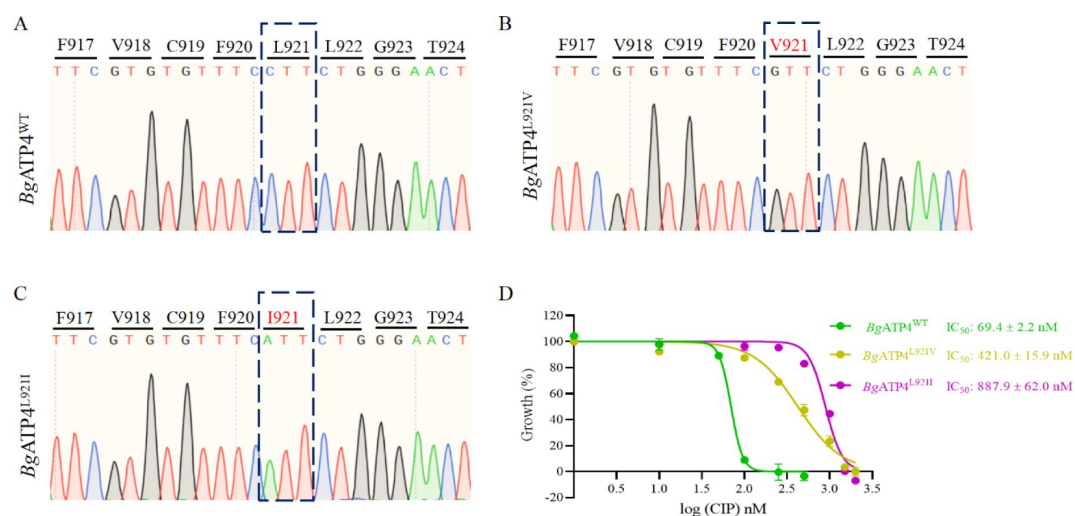
Microscopic observation of thin blood smears was performed to determine the morphological changes of *B. gibsoni* exposed to CIP. The CIP-treated parasites became swollen after incubation with the drug for 72 hr (Figure 3A). For both the treatment and control groups, one hundred parasites were measured. The mean size of treated parasites was notably bigger than the parasites in the untreated group ( $P < 0.0001$ ) (Figure 3B). The addition of the ATP4 inhibitor CIP resulted in a time-dependent increase in the concentrations of  $[Na^+]_i$  in wild-type *B. gibsoni* (Figure 3C), with improved signal-to-noise ratios at the higher drug concentration of 20 nM. We observed that



**Figure 1.**

### CIP demonstrates potent inhibition on *Babesia* spp..

(A) and (B) Dose-dependent growth curve of CIP on *B. bovis* and *B. gibsoni* *in vitro*. Each value represents the mean ± standard deviation (SD) of three independent experiments carried out in triplicate. (C) Inhibitory effects of CIP and atovaquone (ATO) plus azithromycin (AZI) on the proliferation of *B. microti* in BALB/c mice. (D) Hematocrit (HCT) values in mice treated with CIP or ATO plus AZI compared with vehicle-treated mice. (E) Inhibitory effects of CIP and atovaquone (ATO) plus azithromycin (AZI) on the proliferation of *B. rodhaini* in BALB/c mice. (F) Survival rates of CIP-treated, ATO plus AZI-treated, and vehicle-treated mice. The treatment time is shown by arrows, and significant differences ( $P < 0.01$ ) between the drug-treated groups and the vehicle-treated control group are indicated by asterisks. The data from one of six individual experiments are expressed as means ± SD. \*,  $P < 0.05$ ; \*\*,  $P < 0.01$ .



**Figure 2.**

### Mutations in *BgATP4* mediate CIP resistance.

(A-C) Representative sequencing chromatogram of resistant parasites from CIP-treated *B. gibsoni*. The resistant parasite genomic DNA is extracted from blood samples after a 60 day-treatment. The *BgATP4* gene was amplified and sequenced using the DNA. (D) Dose-dependent growth curve of *BgATP4*<sup>WT</sup>, *BgATP4*<sup>L921V</sup>, and *BgATP4*<sup>L921I</sup> *in vitro*. Each value represents the mean ± standard deviation (SD) of three independent experiments carried out in triplicate.

the  $\text{Na}^+$  concentrations in both *BgATP4*<sup>L921V</sup> and *BgATP4*<sup>L921I</sup> lines decreased compared to those of the control *BgATP4*<sup>WT</sup> line after being exposed to 20 nM CIP for 20 min, with a significantly lower  $\text{Na}^+$  concentration in *BgATP4*<sup>L921I</sup> ( $P = 0.0087$ ) (**Figure 3E**). We also demonstrated here that the addition of CIP in wild-type *B. gibsoni* caused an increase in the cytosolic pH (**Figure 3D**). Specifically, 4 min after the drug was added, the average pH of 20 nM CIP group reached as high as 7.278, while the 1 nM CIP group reached 7.089 and the untreated group reached 7.062. The pH values of the 20 nM CIP group were consistently higher than those of the other two groups, although declining with time. In resistant lines, a 20-min exposure to 20 nM CIP caused small changes in the pH values compared to the wild-type line (**Figure 3F**), with the *BgATP4*<sup>L921I</sup> line ( $7.048 \pm 0.042$ ) having a notably lower pH value ( $P = 0.0229$ ).

## Multiple sequence alignment of *Babesia* ATP4 and molecular docking

The whole amino acid sequence of *BgATP4* (GenBank: KAK1443404.1) shared identity values of 29.75%, 49.40%, 49.67%, 62.21%, and 52.47% with *Homo sapiens* ATP4 (GenBank: NM\_000704.3), *P. falciparum* ATP4 (GenBank: PF3D7\_1211900), *T. gondii* ATP4 (GenBank: XP\_018635122.1), *B. bovis* ATP4 (PiroplasmaDB: BBOV\_IV010020), and *B. microti* ATP4 (GenBank: BMR1\_03g01005), respectively (Figure supplement 2).

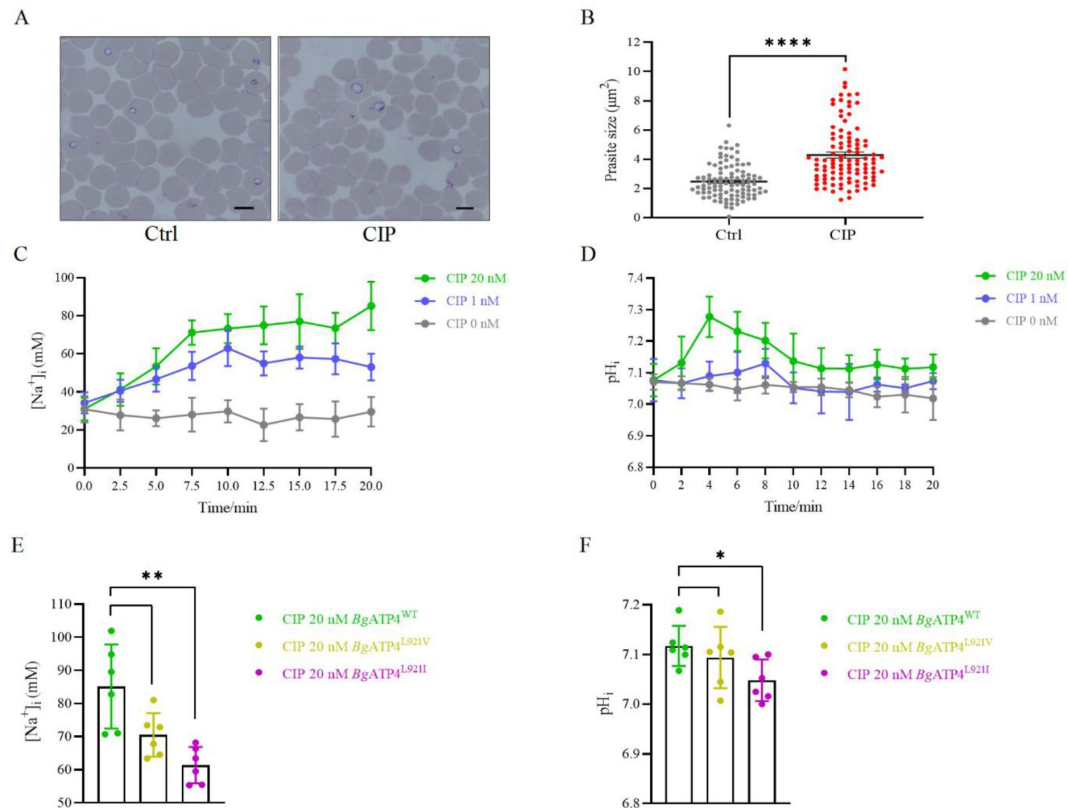
The pLDDT value of *BgATP4*<sup>WT</sup> prediction was 80.7 using Colab-fold. Multiple potential binding sites for CIP were revealed by blind docking throughout the whole protein surface (Figure supplement 3). CIP binds in close proximity to L921, as demonstrated by focused docking on this area (**Figure 4A**). The contribution of each residue to the predicted binding affinity in either mutant structure was reduced; the precise values of *BgATP4*<sup>WT</sup> (**Figure 4B**), *BgATP4*<sup>L921V</sup> (**Figure 4C**) and *BgATP4*<sup>L921I</sup> (**Figure 4D**) were -6.43, -6.40 and -6.26 kcal/mol, respectively. The interactions of CIP from each docking simulation are shown in Table supplement 2.

## Discussion

The repositioning of antimalarial drugs is critical in developing novel strategies for treating babesiosis. CIP is a novel compound that inhibits *Plasmodium* development and has been extensively tested in Phase 1 and Phase 2 clinical trials by targeting *PfATP4* (Qiu et al., 2022). Therefore, our study aimed to repurpose the antimalarial CIP by assessing its efficacy on *Babesia* species. The  $\text{IC}_{50}$  values of CIP against *B. bovis* and *B. gibsoni* *in vitro* were lower than the previously reported  $\text{IC}_{50}$  value of TAF (Ji et al., 2022b) and ATO against *B. gibsoni* (Matsuu et al., 2004). The present investigation also demonstrated that the inhibitory effects of CIP on *B. microti*-infected BALB/c mice were comparable to that of ATO plus AZI, the medication prescribed by the U.S. CDC. CIP was also proven to protect mice from the deadly *B. rodhaini* infection, with protection limited to 66.67% as recorded in the current trial. These results suggest that CIP may be a potential chemotherapy candidate for babesiosis.

In a previous study, the *P. falciparum* Dd2 parent line initially had a subnanomolar  $\text{IC}_{50}$  value, after the CIP-selected cultures to limiting dilution and selected clonal lines, the highly resistant strain reported  $\text{IC}_{50}$  values 4200-4280 fold higher than those of wild-type Dd2 parasites with a G358S mutation in the *PfATP4* protein, a *P. falciparum* P-type  $\text{Na}^+$  ATPase (Qiu et al., 2022). The vital protein ATP4 is found in the parasite plasma membrane and is distinctive to the subclass of the apicomplexan parasite (Mohring et al., 2022). To date, *in vitro* evolution experiments using CIP have produced at least 18 parasite lines with various *PfATP4* mutations (Lee and Fidock, 2016; Rottmann et al., 2010). Another finding showed that the ATP4 activity associated with *TgATP4*<sup>G419S</sup> was 34 times less susceptible to CIP than that of *TgATP4*<sup>WT</sup> (Qiu et al., 2022). It follows that there is a significant chance that the resistant *Babesia* parasites will also emerge from

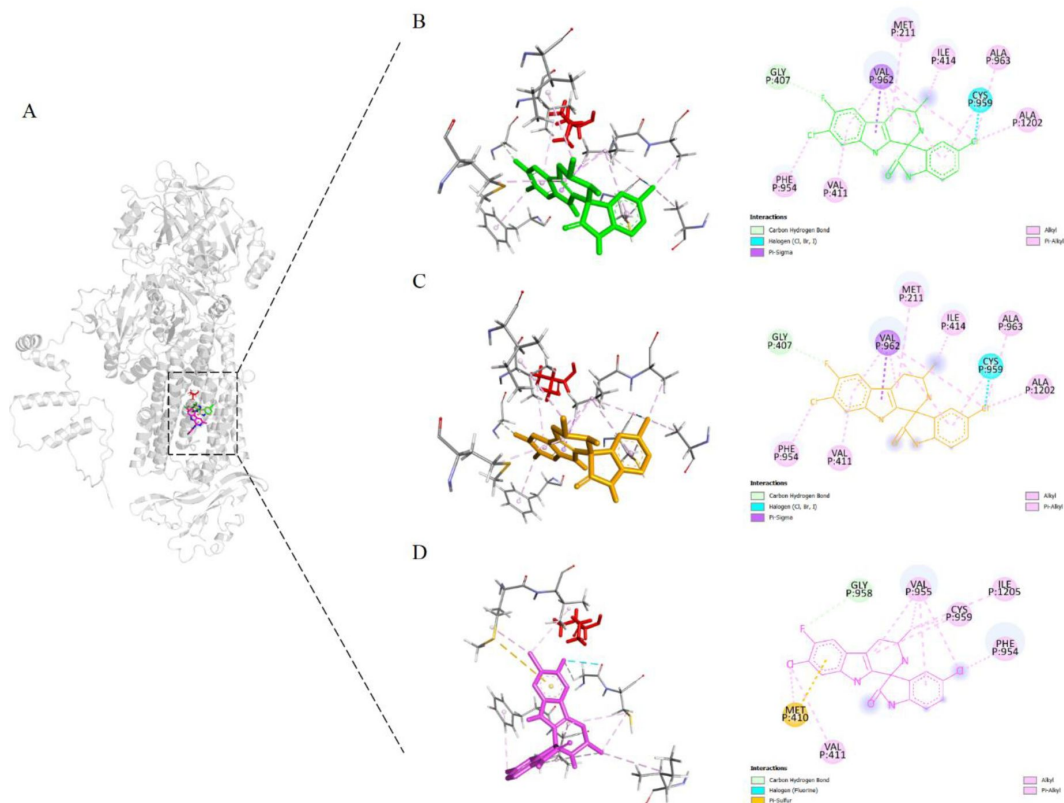




**Figure 3.**

### Effect of CIP on $[Na^+]_i$ and $pH$ regulation in *B. gibsoni* lines.

(A) Untreated and CIP-treated parasite morphology after incubation for 72 h. Scale bar: 5  $\mu m$ . (B) Sizes of 100 parasites in two groups measured with ImageJ software in panels A. Statistically significant differences between the means of variables determined by t-test. \*\*\*\*,  $P < 0.0001$ . (C)  $[Na^+]_i$  concentrations after addition of CIP in *BgATP4*<sup>WT</sup> line. Representative traces from the experiment that highlight the impact of adding 20 nM CIP (green), 1 nM CIP (blue), or 0 nM CIP (grey) on the concentration  $[Na^+]_i$  of the *BgATP4*<sup>WT</sup> line. (D) Alkalinization of  $pH_i$  in *BgATP4*<sup>WT</sup> line upon addition of the ATP4 inhibitor. (E) Addition of 20 nM CIP to the wild-type and resistant parasite lines results in different  $[Na^+]_i$  concentrations. (F) Addition of 20 nM CIP to the wild-type and resistant parasite lines results in different  $pH_i$  concentrations. Experiments were performed in technical duplicates for at least three biological repeats. ns, non significant,  $p > 0.05$ ; \*,  $P < 0.05$ ; \*\*,  $P < 0.01$ .



**Figure 4.**

**Binding sites proximal to *BgATP4* residue 921 predicted by molecular docking.**

(A) The lowest energy poses for CIP were located in reference to the whole protein structure, docking against the WT (green), L921V (yellow), and L921I (pink) mutant *BgATP4*. The side chain of L921 is also shown in a red stick at its position. (B–D) The zoomed views of the binding locations of CIP.



CIP exposure. In this study, we successfully produced two CIP-resistant strains using six independent selections. The newly discovered L921V and L921I mutations in *BgATP4* decreased the CIP sensitivity by 6.1 and 12.8 times, respectively. We provided compelling evidence that natural variety mutations L921V and L921I in *BgATP4* significantly affected the proteins' susceptibility to CIP inhibition, albeit the mutational sites were different from those of *P. falciparum* and *T. gondii*.

Although ATP4 was initially recognized as a  $\text{Ca}^{2+}$  transporter (Krishna et al., 2001 [DOI](#)), current evidence suggests that ATP4 functions as an ATPase for exporting  $\text{Na}^+$  while importing  $\text{H}^+$  (Mohring et al., 2022 [DOI](#)), as well as causing a variety of other physiological perturbations including an increase in the volume of parasites and infected erythrocytes, due to the osmotic impact of the  $[\text{Na}^+]_{\text{cyt}}$  increase (Dennis et al., 2018 [DOI](#)), a decline in cholesterol extrusion from the parasite plasma membrane as a result of the increase in  $[\text{Na}^+]_{\text{cyt}}$  (Das et al., 2016 [DOI](#)), and an intensified rigidity of erythrocytes infected with ring-stage parasites (Zhang et al., 2016 [DOI](#)). In this study, the CIP-exposed wild-type *B. gibsoni* became swollen, which was identical to a prior study on *P. falciparum* (Dennis et al., 2018 [DOI](#)). We hypothesize that the swelling of the isolated parasites can be ascribed to the osmotic consequences of  $\text{Na}^+$  uptake and is contingent upon the presence of  $\text{Na}^+$  in the external environment. This hypothesis is supported by the following experimental results.

The output of  $\text{Na}^+$  and input of  $\text{H}^+$  diminished upon CIP-induced inhibition of *PfATP4*, and the continued outflow of  $\text{H}^+$  via V-type  $\text{H}^+$ -ATPase led to an alkalinization that ultimately killed the parasites (Spillman et al., 2013 [DOI](#)). Our capacity to measure a time-dependent increase in the concentration of  $[\text{Na}^+]_{\text{i}}$  and pH value for wild-type *B. gibsoni* facilitated us to gain insight into the basic mechanisms of CIP on *BgATP4*<sup>WT</sup> function. Furthermore, our results validate that internal alkalinization was the main factor in *Babesia* death and support our hypothesis that the swollen isolated parasites were produced by  $\text{Na}^+$  absorption. To explore further how mutations in *BgATP4* are associated with the upregulation of the parasite's  $[\text{Na}^+]_{\text{i}}$  and  $[\text{H}^+]_{\text{i}}$ , we tested two *BgATP4*-mutant lines that were chosen previously with *BgATP4* inhibitors (*BgATP4*<sup>L921V</sup> and *BgATP4*<sup>L921I</sup>). Due to the presence of L921V and L921I mutations in drug-resistant strains of *BgATP4*, the concentration of  $\text{Na}^+$  did not increase as much as it would have in the wild-type strain following the addition of 20 nM CIP for 20 min. As intraerythrocytic alkalinization rises, the same outcome happens. From these results, we deduced how natural mutations in *BgATP4* may affect ATP4 inhibitor susceptibility by dysregulating  $\text{H}^+$  and  $\text{Na}^+$  balance, which helped parasites survive in a relatively high concentration of CIP. An illustration of the putative processes for regulating  $\text{Na}^+$  and  $\text{H}^+$  in erythrocytes infected with the *Babesia* parasite is presented in Figure supplement 4.

According to studies using the *PfATP4* model for molecular docking, the G358S mutation results in a steric clash that lowers CIP's binding affinity (Qiu et al., 2022 [DOI](#)). The results from the current study were similar to the *PfATP4* model. The molecular docking was constructed using a ColabFold model of wild-type *BgATP4*, which predicted the binding mode and affinity between ATP4 protein and the ligand to provide a possible mechanistic explanation. It suggested that the L921V mutation caused changes at the atomic level, whereas the L921I mutation created a steric clash that reduced the binding affinity of CIP. The predicted affinity score of the L921I mutation was lower than that of the L921V mutation. Thus, it is possible that *BgATP4*<sup>L921I</sup> has a weaker binding to the drug compared to *BgATP4*<sup>L921V</sup>. These findings are consistent with the results obtained from measuring the IC<sub>50</sub> of the drug against parasites with the L921V and L921I mutations.

## Materials and methods

## Parasite culture

The parasites *B. gibsoni* (Oita strain) and *B. bovis* Texas strain were *in vitro* cultured in 24-well plates and maintained in an atmosphere of 5% CO<sub>2</sub> and 5% O<sub>2</sub> at 37 °C (Liu et al., 2018 [DOI](#)). For the *in vivo* studies, *B. microti* Peabody mjr strain-(ATCC PRA-99) and *B. rodhaini* Australia strain-infected RBCs (iRBCs), which were collected and diluted with phosphate-buffered saline (PBS) when the parasitemia levels in the donor mice reached ~20% and 50%, respectively, and were intraperitoneally injected into BALB/c mice. Each BALB/c mouse was infected with  $1.0 \times 10^7$  *B. microti* or *B. rodhaini* iRBCs for the *in vivo* trials (Ji et al., 2022a).

## *In vitro* cytotoxicity of CIP in canine cells and hemolysis rate in canine erythrocytes

Cultures of Madin-Darby canine kidney (MDCK) cells were maintained at 37 °C under an atmosphere of 5% CO<sub>2</sub> and 5% O<sub>2</sub> and the cytotoxic effect of CIP (MedChem Express, Tokyo, Japan) was assessed using a cell viability assay by CCK-8 (Dojindo, Japan) as described previously (Li et al., 2023 [DOI](#)). The selectivity index is calculated as the ratio between the half maximal inhibitory concentration (IC<sub>50</sub>) and the 50% cytotoxic concentration (CC<sub>50</sub>) values. Canine erythrocytes were collected from healthy beagle dogs raised in NRCPD and stocked in Vega y Martinez (VYM) phosphate-buffered saline solution at 4 °C (Vega et al., 1985 [DOI](#)). A canine erythrocyte hemolysis assay was performed at concentrations of 0.1, 1, 5, 10, 25, 50, and 100 µM as previously described (Ariefte et al., 2022 [DOI](#)).

## Evaluation of the efficacy of CIP against *Babesia* parasites *in vitro*

The efficacy of CIP against *B. gibsoni* and *B. bovis* was determined using a fluorescence assay, as previously described (Guswanto et al., 2014 [DOI](#)). The IC<sub>50</sub> values were determined from the fluorescence values and by non-linear regression analysis (curve fit) in GraphPad Prism 9 (GraphPad Software Inc., USA).

## Chemotherapeutic effects of CIP against *Babesia* infections *in vivo*

CIP was evaluated on *B. microti*- and *B. rodhaini*-infected mice, as previously described (Ndayisaba et al., 2021 [DOI](#); Tuvshintulga et al., 2022 [DOI](#)). When all mice had a 1% average parasitemia at 4 days postinfection (DPI), the drug treatments were administered, which continued for seven days. Three groups of *B. microti*-infected mice were administered different treatments. The CIP group (n = 6) and the ATO plus AZI group (n = 6) were orally treated with 20 mg/kg CIP and 20 mg/kg ATO plus AZI (Sigma, Tokyo, Japan), respectively. The infected mice of the vehicle group (n = 6) orally received 0.2 mL of sesame oil as the control. Eighteen mice infected with *B. rodhaini* were likewise placed into three groups for treatment, and they received the same care as the mice infected with *B. microti*. A light microscope (Nikon, Japan) and a hematology analyzer (Celltac α MEK-6450, Nihon Kohden Corporation, Tokyo, Japan) were used to observe the parasitemia and hematocrit levels every two and four days, respectively.

## Selection of CIP-resistant *B. gibsoni* *in vitro*

Selections were initiated by exposing six independent flasks, each containing 10 µL *B. gibsoni* iRBCs mixed with 40 µL RBC into a 450 µL culture medium, which contained increasing concentrations of CIP: 5, 10, 20, 30 to 694 nM (10×IC<sub>50</sub>). The medium containing CIP was replaced daily until parasites treated with 10×IC<sub>50</sub> CIP reached multiplication rates that were approximately comparable to those of the untreated controls (Hwang et al., 2010 [DOI](#)). Then, to evaluate the decreased sensitivity to CIP, IC<sub>50</sub> values of resistant strains were determined by nonlinear regression using the GraphPad Prism software.

## Detection of *B. gibsoni* ATP4 gene mutations

The genomic DNA of the mutant parasites was extracted and sequenced (Ji et al., 2022a). The primer sets used for sequencing are listed in Table supplement 1. The single nucleotide variants were confirmed by pairwise alignment to the *BgATP4*<sup>WT</sup> sequence (GenBank: JAVEPI010000002.1).

## Morphological changes in CIP-treated *in vitro* cultured *B. gibsoni*

A microscopy assay was used to detect the morphological changes of wild-type *B. gibsoni* after exposure to 50 nM CIP for three consecutive days (Liu et al., 2021; Tayebwa et al., 2018). At 4 days post-treatment, ImageJ software was used to measure the sizes of 100 randomly selected parasites in the CIP-treated group and the control group on Giemsa-stained blood smears.

## Parasites [Na<sup>+</sup>]<sub>i</sub> and pH<sub>i</sub> measurements

For both [Na<sup>+</sup>]<sub>i</sub> and pH<sub>i</sub> measurements, the wild-type and mutant parasites were initially separated from erythrocytes by treatment with saponin (0.05% [wt/vol]) for 5 min (Saliba and Kirk, 1999). The Na<sup>+</sup>-sensitive fluorescent dye SBFI (Thermo Fisher Scientific; product S1263) was used to quantify intracellular sodium ([Na<sup>+</sup>]<sub>i</sub>). Saponin-isolated parasites (at 1×10<sup>8</sup> parasites/mL) were loaded with SBFI (5 μM; in the presence of 0.02% w/v Pluronic F127) for 1 hr at room temperature (RT) (Spillman et al., 2013). Thereafter, SBFI-loaded parasites were resuspended in physiological saline (120 mM NaCl, 5 mM KCl, 25 mM HEPES, 20 mM D-glucose, and 1 mM MgCl<sub>2</sub> [pH 7.1]) at RT in the presence or absence of CIP. A 96-well microtiter plate was filled with around 200 μL of parasite suspension per well. The dye-loaded cells fluoresced at 515 nm after being stimulated at 340 and 380 nm. Parasites loaded with SBFI were suspended in calibration buffers containing [Na<sup>+</sup>] values ranging from 0 to 140 mM (pH 7.1) to establish calibration curves (Diarra et al., 2001).

The cytosolic pH of wild-type and mutant strains was measured using the pH-sensitive fluorescent dye BCECF [2',7'-bis-(2-carboxyethyl)-5-(and-6)-carboxyfluorescein] (Biotium; product 51011). The BCECF was added to saponin-isolated parasites by suspension (1×10<sup>8</sup> parasites/mL) and incubated for 20 minutes at 37 °C in RPMI-1640 culture medium (Gibco, USA) (Saliba and Kirk, 1999). Thereafter, the parasites loaded with dye were rinsed thrice (12,000 × g, 1 min) in RPMI-1640 culture medium, then resuspended in physiological saline (as previously mentioned) with or without various concentrations of CIP. A 96-well microtiter plate was filled with around 200 μL of parasite suspension per well. The dye-loaded cells fluoresced at 520 nm after being stimulated at 440 and 490 nm. A pH calibration was carried out for each experiment (Mohring et al., 2022).

## Multiple ATP4 sequence alignment and molecular docking

*B. gibsoni* ATP4 (GenBank: KAK1443404.1), *B. bovis* ATP4 (PiroplasmaDB: BBOV\_IV010020), *B. microti* ATP4 (GenBank: BMR1\_03g01005), *T. gondii* ATP4 (GenBank: XP\_018635122.1), and *Homo sapiens* ATP4 (GenBank: NM\_000704.3) sequences were obtained by a homology search using *P. falciparum* ATP4 (GenBank: PF3D7\_1211900). Sequence alignment was analyzed using MUSCLE in Jalview v2.11.3.2 software and BLAST (<http://www.ncbi.nlm.nih.gov/BLAST/>).

AlphaFold was used to predict the structure of *BgATP4*<sup>WT</sup> (Jumper et al., 2021). Using the GROMACS 2021 Molecular Dynamics package, the energy minimization was carried out following the model's generation (Lindahl et al., 2021). The mutations were produced by using Charmm-GUI PDB reader (Jo et al., 2014). PyMol (version 2.0 Schrödinger, LLC) was used to confirm the position of the mutation site (center: -20.367, 10.904, 9.435; size: 30×30×30) (Trott and Olson, 2010). The ligand molecules CIP was downloaded from PubChem (CID 44469321 (<https://pubchem.ncbi.nlm.nih.gov/compound/44469321>)). The ligand was used to dock with Gnina

(Eberhardt et al., 2021 [DOI](#)). The affinity score and binding pose were chosen only from the highest CNN (convolutional neural network) score results from docking simulations. The models were visualized using the PyMOL Molecular Graphic System and Discovery Studio.

## Statistical analysis

Data analysis, namely one-way analysis of variance (ANOVA) and two-tailed unpaired t-tests, was performed using GraphPad Prism (La Jolla, CA, USA) version 9. A *P* value of <0.05 was considered a statistically significant result.

## Competing interest

The authors have declared that no conflict of interest exists.

## Acknowledgements

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## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

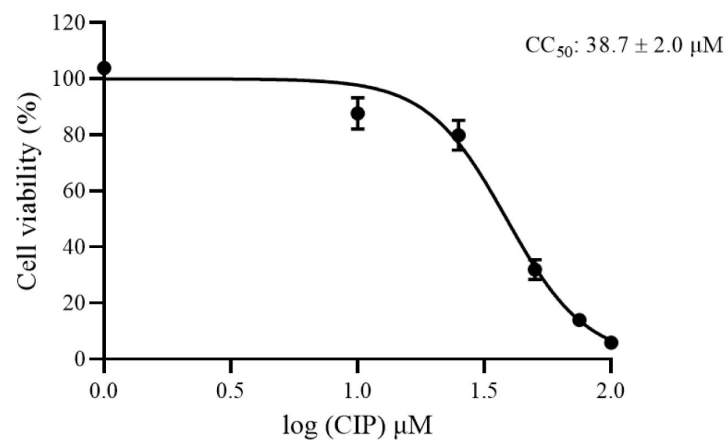
## Author contributions

Hang Li, Investigation, Data curation, Formal analysis, Writing – original draft, Writing – review & editing; Shengwei Ji, Methodology, Investigation, Software, Validation; Nanang R. Ariefta, Methodology, Software, Formal analysis; Eloiza May S. Galon, Visualization, Writing – review & editing; Shima AES El-Sayed, Methodology, Visualization; Lijun Jia, Data curation, Visualization; Yoshifumi Nishikawa, Methodology, Resources; Mingming Liu, Conceptualization, Methodology, Investigation, Resources, Data curation, Funding acquisition, Supervision, Writing – review & editing; Xuenan Xuan, Conceptualization, Methodology, Validation, Formal analysis, Supervision, Project administration, Writing – review & editing.

## Ethics

All animal experiments were performed according to the Guide for the Care and Use of Laboratory Animals of Obihiro University of Agriculture and Veterinary Medicine, which was also approved by the Committee on the Ethics of Animal Experiments at the Obihiro University of Agriculture and Veterinary Medicine, Japan (permit numbers: animal experiment, 22-145 and 23-132; DNA experiment, 2207 and 2208; pathogen, 202308 and 202306).

## Supplementary Material



**Figure supplement 1.**

**Cytotoxicity assay of CIP on the MDCK cell line.**

The MTP-500 microplate reader is utilized to detect the absorbance at 450 nm. The results are displayed as the mean  $\pm$  standard deviation of three separate tests.



*Homo sapiens*/1-1035  
*P. falciparum*/1-1264  
*T. gondii*/1-1338  
*B. gibsoni*/1-1264  
*B. bovis*/1-1261  
*B. microti*/1-1234

60 .....AELEKGYTSATKQLSFLAAALLLRD.....SPHAPPPPGTGYVNHARLAGGLQGLWVAAALCLIFAIQASE  
 60 AOHGDESPRRDSSHLLRRFSSKRESTVGGSGTGHSQLOKSISSVQSMRDEHARLLPASSMSLAPQMGYENYETVIDRLKSHVSLKELVETARQSDPE  
 60 .....ITSFSA.....DNVSESEATRLSKHLNHAEDFIEVARMDSL  
 60 .....DMVORLSLNQLSPREGVIEWSKVDNDE

*Homo sapiens*/1-1035  
*P. falciparum*/1-1264  
*T. gondii*/1-1338  
*B. gibsoni*/1-1264  
*B. bovis*/1-1261  
*B. microti*/1-1234

104 IIVK.....EESPKSIDYNYASESIENLCKEFGLESINTOLNSEGVKINRDKYGENFIEKGEVVPVLLILSDYGCSPVULLLLVAATASVAL.....  
 100 AFRAVVTLEQSPHTSKNRRFASLIEELAKEFGKMDSTOLTTEEVLENRRRYGPHVLEKDKSEPVKKIIGQFLSPVULLLLVAATASVAL.....  
 96 MVEKAKKIKLEGRPSRSHVYTAIEVIMOGFGLNDLTGLTKACDELHEELYGRNILEASRKLPRWVAAAFANFVULLLSAAIASAL.....  
 96 CFMIARESAQGGPHTKSLFTDTIENIMKVGLNSLDTGLDEVEQNRKLYGENAINVQKSESIKIKISYTYTFNFIALLIAAIVLAL.....

*Homo sapiens*/1-1035  
*P. falciparum*/1-1264  
*T. gondii*/1-1338  
*B. gibsoni*/1-1264  
*B. bovis*/1-1261  
*B. microti*/1-1234

133 GDLTTDDLLYLALIAVAVVVTGCGVYGFKSTNIISFKNLVPOOATVIRQDQKFOHADQLVVLQVEMKQDGRMPAIIRLAAGCGVDNSLTGE  
 192 .....NEVVEGVAIIISIVTLNACLATYMEKSSQDAIOKLAMASGCTVLRNGQKVVPREVVVGGVLLINTQSSISADLRFDFVIELKTNESLLTGE  
 253 .....QEWVEGAAFIIVTLNACLATYMEKSSQDAIOKLAMASGCTVLRNGQKVVPREVVVGGVLLINTQSSISADLRFDFVIELKTNESLLTGE  
 189 .....GNVIEGAFIIVTINAMQATYMEKSSADALEKLASISATTTVINGEIEIDSKDQVGGVILNMDSTPADVRIEVEKILNHEALLTGE  
 179 .....GNVIEGAFIIVTINAMQATYMEKSSADALEKLASISATTTVINGEIEIDSKDQVGGVILNMDSTPADVRIEVEKILNHEALLTGE

*Homo sapiens*/1-1035  
*P. falciparum*/1-1264  
*T. gondii*/1-1338  
*B. gibsoni*/1-1264  
*B. bovis*/1-1261  
*B. microti*/1-1234

233 EPQPTSPPECTHESPLETRIAFFTMCLERIVQDLVNTQDRIITIKIASL--ASQVENEKTRIAIEIEHFVDIADQLLFGATFFIAMCIGIT--  
 286 SEDIKRTIVADNLESTPATNLCPATISVTSQSGKQIVISTOLDTVQVKIASQKKSSKQKLTPLQVALNKKILQIGLITAIIVLVVITISLRVIAIKVRDA  
 347 SEDIKRTIVADNLESTPATNLCPATISVTSQSGKQIVISTOLDTVQVKIASQKKSSKQKLTPLQVALNKKILQIGLITAIIVLVVITISLRVIAIKVRDA  
 283 EDVIRKQLVADGLDSTDLKQFAMTIEIASASAKAVAVISTOLDTVQVKIASQKKSSKQKLTPLQVALNKKILQIGLITAIIVLVVITISLRVIAIKVRDA  
 283 SEPDKHLVADGLDGLNAPFATMCFASTSVVCSAKQIVVITOMOTVQVGIATOLKSALSGSDVTPLOGLSSNSGLIIGIISIFVLITIVVAIVTVEDPT  
 273 SEDVKKITLLPDPDPSSSLLVFSSTGVNMGKAIIVTHVMOTVQVKIASQKKSSKQKLTPLQVALNKKILQIGLITAIIVLVVITISLRVIAIKVRDA

*Homo sapiens*/1-1035  
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*B. gibsoni*/1-1264  
*B. bovis*/1-1261  
*B. microti*/1-1234

328 .....FIRAMVFFMAIVAYVPEGLLATVTVLSLTKRLASKNCVKKNEIARVETLQSTVICSDKTGTLTQNRMTYSHLW.....  
 386 HADKDPFTVITIGVGFVAVSSIEPGLPMVVTISARAKDMVKKNAANVRKLPVAVETLQSTVICSDKTGTLTQNRMTYSHLW.....  
 447 HADKDPFTVITIGVGFVAVSSIEPGLPMVVTISARAKDMVKKNAANVRKLPVAVETLQSTVICSDKTGTLTQNRMTYSHLW.....  
 383 RPDTRNVLLAVLLAVGFVAVSIEPGLPMVVTISARAKDMVKKNAANVRKLPVAVETLQSTVICSDKTGTLTQNRMTYSHLW.....  
 373 HNDNRILTLILAVGFVAVSSIEPGLPMVVTISARAKDMVKKNAANVRKLPVAVETLQSTVICSDKTGTLTQNRMTYSHLW.....

*Homo sapiens*/1-1035  
*P. falciparum*/1-1264  
*T. gondii*/1-1338  
*B. gibsoni*/1-1264  
*B. bovis*/1-1261  
*B. microti*/1-1234

404 .....DNHNIHADITED.....GSG--GTGDSSETWRA.....LCRVLTLENRAAFKSGQDAV.....PVPKRIVIGDASATL  
 486 FDVFPKVFESGLDNLSEKKEIKVIAKNQ--TSYQKLVNYGNPSKSEV--IVDKTRSLMFAAVLNVDYTTLSRDPKTKLKWIKHNSMGPI  
 546 FDEVFYKVFESGLDNLSEKKEIKVIAKNQ--TSYQKLVNYGNPSKSEV--IVDKTRSLMFAAVLNVDYTTLSRDPKTKLKWIKHNSMGPI  
 478 LFLFPLDGFNFPFQKKNLGGADILLEMAKQ--VPIQAKYKRLCSAKKGA--VERQKAGELISAVLNRSQKIERENHTDQKGVNHTCEPL  
 478 LFLFPLDGFNFPFQKKNLGGADILLEMAKQ--VPIQAKYKRLCSAKKGA--VERQKAGELISAVLNRSQKIERENHTDQKGVNHTCEPL  
 471 KEFPLDGFNFPFQKKNLGGADILLEMAKQ--VPIQAKYKRLCSAKKGA--VERQKAGELISAVLNRSQKIERENHTDQKGVNHTCEPL

*Homo sapiens*/1-1035  
*P. falciparum*/1-1264  
*T. gondii*/1-1338  
*B. gibsoni*/1-1264  
*B. bovis*/1-1261  
*B. microti*/1-1234

468 L.....KFSLETLGNAMG.....YDRFEEKVC--EIPFNSTNIFQLSITLIE.....DPRDPRHLVVKAGAPRVLERO--SSILIKQ--  
 581 VAAAKIVSYSEINFNHKS.....YDRFEEKVC--EIPFNSTNIFQLSITLIE.....DPRDPRHLVVKAGAPRVLERO--SSILIKQ--  
 573 VAAAKIVSYSEINFNHKS.....YDRFEEKVC--EIPFNSTNIFQLSITLIE.....DPRDPRHLVVKAGAPRVLERO--SSILIKQ--  
 573 VAAAKIVSYSEINFNHKS.....YDRFEEKVC--EIPFNSTNIFQLSITLIE.....DPRDPRHLVVKAGAPRVLERO--SSILIKQ--  
 567 VAAAKIVSYSEINFNHKS.....YDRFEEKVC--EIPFNSTNIFQLSITLIE.....DPRDPRHLVVKAGAPRVLERO--SSILIKQ--

*Homo sapiens*/1-1035  
*P. falciparum*/1-1264  
*T. gondii*/1-1338  
*B. gibsoni*/1-1264  
*B. bovis*/1-1261  
*B. microti*/1-1234

538 .....ELPLDEQWREAFQTAYLSGGGLGRLVGLFCOLYNEKYPP.....GYAFDVEAMNFPSSQCFALVSMIDPPRATIPDVLKRT--  
 674 VQVSW--NSTITQERNVLIKKLELQKALRVLICIKPLTDQNIIEELKKLEBAREKLYVN--YDENGOFIPMYVASFPPRPPKTKLKWIKHNSMGPI  
 740 PSVEW--EKOMTPEEIMKVEAVNLELQKALRVLICIKPLTDQNIIEELKKLEBAREKLYVN--YDENGOFIPMYVASFPPRPPKTKLKWIKHNSMGPI  
 664 CTIDWHM--DINESKIMEKFLKNDLSSRALRVLVGMFLTDVTSVESKRCDDSRLLWLLTGKMGSGCLVLLFISGLDPPFPFQKGVNHTCEPL  
 664 CVIDWASQ--IENNETLLNVMKANQDSSRALRVLVGMFLTDVTSVESKRCDDSRLLWLLTGKMGSGCLVLLFISGLDPPFPFQKGVNHTCEPL  
 653 C.....HMGVDNLLAKIMSDANSLSKNAYVLLLAICPLIAEDLENLERLESSD--KOEYIL.....GKKIIVLRLTSGVDPSPVSKKAVVSGESA

*Homo sapiens*/1-1035  
*P. falciparum*/1-1264  
*T. gondii*/1-1338  
*B. gibsoni*/1-1264  
*B. bovis*/1-1261  
*B. microti*/1-1234

621 IIRVIMVTGGDPITAKATAASVGIISSEGTVEPIAARLRVPVDDVNRKQARACVINGMOLKMDPSELVEALRHPMEVFARTSGOKLVIVESCOLRL  
 768 GVRVIMVTGGDPITAKATAASVGIISSEGTVEPIAARLRVPVDDVNRKQARACVINGMOLKMDPSELVEALRHPMEVFARTSGOKLVIVESCOLRL  
 834 GVRVIMVTGGDPITAKATAASVGIISSEGTVEPIAARLRVPVDDVNRKQARACVINGMOLKMDPSELVEALRHPMEVFARTSGOKLVIVESCOLRL  
 768 GVRVIMVTGGDPITAKATAASVGIISSEGTVEPIAARLRVPVDDVNRKQARACVINGMOLKMDPSELVEALRHPMEVFARTSGOKLVIVESCOLRL  
 764 GVRVIMVTGGDPITAKATAASVGIISSEGTVEPIAARLRVPVDDVNRKQARACVINGMOLKMDPSELVEALRHPMEVFARTSGOKLVIVESCOLRL  
 739 GVRVIMVTGGDPITAKATAASVGIISSEGTVEPIAARLRVPVDDVNRKQARACVINGMOLKMDPSELVEALRHPMEVFARTSGOKLVIVESCOLRL

*Homo sapiens*/1-1035  
*P. falciparum*/1-1264  
*T. gondii*/1-1338  
*B. gibsoni*/1-1264  
*B. bovis*/1-1261  
*B. microti*/1-1234

721 AIVRVITGGVNDGAPALKAADISVAMGIASTDAIKANADILLDDNFASITGVGGRLFDRLKNSIAYTITKNTILPTLYLIYITSVFPLGCGITLFL  
 858 YLVAMTGGVNDGAPALKAADISVAMGIASTDAIKANADILLDDNFASITGVGGRLFDRLKNSIAYTITKNTILPTLYLIYITSVFPLGCGITLFL  
 920 HTVAMTGGVNDGAPALKAADISVAMGIASTDAIKANADILLDDNFASITGVGGRLFDRLKNSIAYTITKNTILPTLYLIYITSVFPLGCGITLFL  
 852 YLVAMTGGVNDGAPALKAADISVAMGIASTDAIKANADILLDDNFASITGVGGRLFDRLKNSIAYTITKNTILPTLYLIYITSVFPLGCGITLFL  
 853 YLVAMTGGVNDGAPALKAADISVAMGIASTDAIKANADILLDDNFASITGVGGRLFDRLKNSIAYTITKNTILPTLYLIYITSVFPLGCGITLFL  
 823 YLVAMTGGVNDGAPALKAADISVAMGIASTDAIKANADILLDDNFASITGVGGRLFDRLKNSIAYTITKNTILPTLYLIYITSVFPLGCGITLFL

*Homo sapiens*/1-1035  
*P. falciparum*/1-1264  
*T. gondii*/1-1338  
*B. gibsoni*/1-1264  
*B. bovis*/1-1261  
*B. microti*/1-1234

921 IELCTIIFSGAVKAEQDIHLPRFNRPRDLVNE.....LAAY.....SYFGIGATOS.....FA  
 958 LNLMDGCPAVALSREPREDDNNKTPPRPKKQIMFKRWNYGLLPHITFEALCILLSLAFSLVICTGTYNLNGLNLSKTNLNDVNDANVYVHYKQV  
 1020 LNLMDGCPAVALSREPREDDNNKTPPRPKKQIMFKRWNYGLLPHITFEALCILLSLAFSLVICTGTYNLNGLNLSKTNLNDVNDANVYVHYKQV  
 962 LNLMDGCPAVALSREPREDDNNKTPPRPKKQIMFKRWNYGLLPHITFEALCILLSLAFSLVICTGTYNLNGLNLSKTNLNDVNDANVYVHYKQV  
 963 LNLMDGCPAVALSREPREDDNNKTPPRPKKQIMFKRWNYGLLPHITFEALCILLSLAFSLVICTGTYNLNGLNLSKTNLNDVNDANVYVHYKQV

*Homo sapiens*/1-1035  
*P. falciparum*/1-1264  
*T. gondii*/1-1338  
*B. gibsoni*/1-1264  
*B. bovis*/1-1261  
*B. microti*/1-1234

876 GFTDFTTAMQEDWF.....FLLCVGLRAQWED.....HHLOBLDQSYGQEW.....  
 1058 SYEYFISTDVGVWTVNVSFWDPQNNNAEYVFWAAKQVNIHRLSDIVHPELRRLMQD.....GCSQBLTLBENCRPKPKNTSDQYND.....ELE  
 1119 RYEVYVTDITBWVTHIDFWNRKTKMEQVGLAALAKHPHVTIVETPOLAKYIVEMASS.....GCPEDVDTSTETFCMPKAGTKVSSATD.....TFK  
 1050 SYEYVGVNINIDWVTNIDYDPEDEKMKMTLGAAGRVYLLRPSAEIHQEIKNFSIA.....SKLVVIGOVENDEMMYRQKSGVVPKQKNGVYQRI  
 1051 SYEYVGVNINIDWVTNIDYDPEDEKMKMTLGAAGRVYLLRPSAEIHQEIKNFSIA.....SKLVVIGOVENDEMMYRQKSGVVPKQKNGVYQRI  
 1020 SYEYVGVNINIDWVTNIDYDPEDEKMKMTLGAAGRVYLLRPSAEIHQEIKNFSIA.....SKLVVIGOVENDEMMYRQKSGVVPKQKNGVYQRI

*Homo sapiens*/1-1035  
*P. falciparum*/1-1264  
*T. gondii*/1-1338  
*B. gibsoni*/1-1264  
*B. bovis*/1-1261  
*B. microti*/1-1234

919 .....FGQRLYQDYTGTVFISIEVQIADVLIRKTLRLSAFOGGRNKLIVLIVFGVQIGCFGLVGFMPNIFNMFIRFQW.....VLPLVGLI  
 1147 LILKKGEDVITAKGSKRGRTMAISAWCEMLRAY.....TVRSWEYFYKFNRRNPMWHLACSISATLITSLTICIPGITSILNTTCLL--WQVYLLAIGQGFV  
 1200 ISAPKQYFDVSARQAKMORTCSFITAVWCEMLRAY.....TVRSWEYFYKFNRRNPMWHLACSISATLITSLTIVPGIQSASTCALP--WQVYLLAIGQGFV  
 1147 ISAPKQYFDVSARQAKMORTCSFITAVWCEMLRAY.....TVRSWEYFYKFNRRNPMWHLACSISATLITSLTIVPGIQSASTCALP--WQVYLLAIGQGFV  
 1147 ISAPKQYFDVSARQAKMORTCSFITAVWCEMLRAY.....TVRSWEYFYKFNRRNPMWHLACSISATLITSLTIVPGIQSASTCALP--WQVYLLAIGQGFV  
 1115 ISAPKQYFDVSARQAKMORTCSFITAVWCEMLRAY.....TVRSWEYFYKFNRRNPMWHLACSISATLITSLTIVPGIQSASTCALP--WQVYLLAIGQGFV

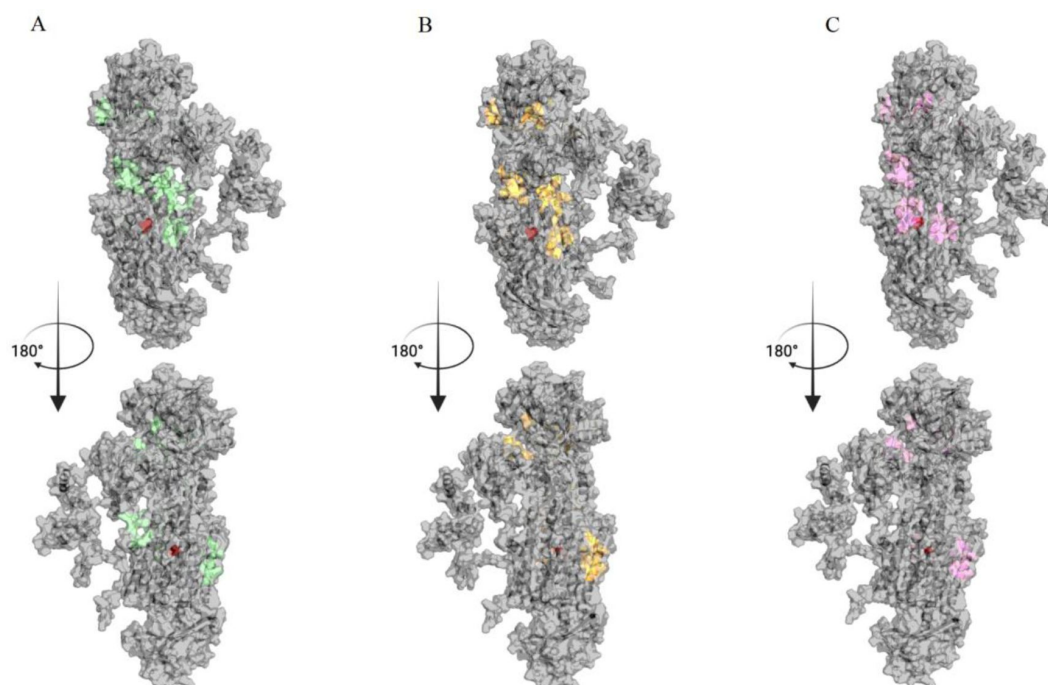
*Homo sapiens*/1-1035  
*P. falciparum*/1-1264  
*T. gondii*/1-1338  
*B. gibsoni*/1-1264  
*B. bovis*/1-1261  
*B. microti*/1-1234

1010 IFVYDEIRKLQVRCCFQSWDQELVY.....  
 1243 NLFDDEL.....VPRVIVRRNYMTIKN.....  
 1305 NLFDDEL.....VPRVIVRRNYMTIKN.....  
 1243 NLFDDEL.....VPRVIVRRNYMTIKN.....  
 1243 NLFDDEL.....VPRVIVRRNYMTIKN.....

Figure supplement 2.

## Multiple sequence alignment of ATP4 in different species.

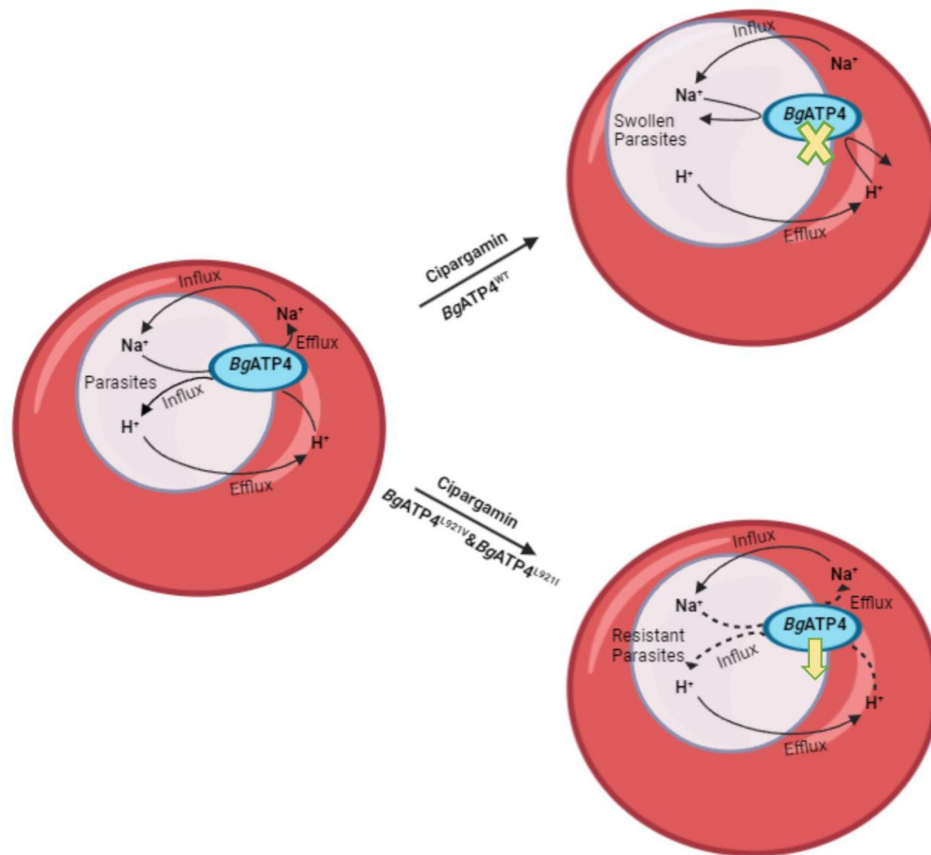
A yellow square denotes the *Bg*ATP4 mutation site discovered in this investigation; red squares represent sites linked to *P. falciparum* CIP resistance, and a green square represents sites associated with *T. gondii*.



**Figure supplement 3.**

**Binding sites for CIP found by Gnina search across the entire surface of the protein.**

(A-C) The binding space of WT, L921V, and L921I mutants in *Bg*ATP4 are labeled in green, yellow and pink, respectively.



**Figure supplement 4.**

**Proposed mechanism of inhibition of CIP on wild-type and mutant parasite-infected erythrocytes.**

CIP disrupts the *BgATP4* function of wild-type parasites, which causes a net influx of  $\text{Na}^+$  and efflux of  $\text{H}^+$  from the parasite. The osmotic load imposed on the influx of  $\text{Na}^+$  further brings about parasite swelling and internal alkalization, which are the main factors in *Babesia* death. Mutations in ATP4 minimize the susceptibility to ATP4 inhibitors by recovering  $\text{H}^+$  and  $\text{Na}^+$  balance.

| Gene                          | Primer (5'-3')                   |
|-------------------------------|----------------------------------|
| <i>Bg</i> ATP4 1F             | ACGTAGTTGTCGAACATCTTACACCAATGC   |
| <i>Bg</i> ATP4 1R             | CAGAGGTAGCATAGTGAGATCGCCCCGA     |
| <i>Bg</i> ATP4 2F             | GAGATTGCAAGGTTTACGGATGAGTCTATGG  |
| <i>B<sup>g</sup></i> ATP4 2R  | ATGACAACGTCACCGGGTACAACACGCTTG   |
| <i>Bg</i> ATP4 3F             | GAAAAGCTTGCTCAAATGTCTTCACCGACAAC |
| <i>B<sup>g</sup></i> ATP4 3R  | GTTTCTGGACGAGTTGGATCCTCATAACC    |
| <i>Bg</i> ATP4 4F             | AAATTGGGGGGGATAATTGGCATTATCTCC   |
| <i>B<sup>g</sup></i> ATP4 4R  | ACCTTATCAAAGGGTACCTCTTGCTTTG     |
| <i>Bg</i> ATP4 5F             | TCAATCCATTTCGGTGAATTTTCAAAAAGG   |
| <i>B<sup>g</sup></i> ATP4 5R  | GTGTTTGAATAACATATCTGGTGCCCCCTT   |
| <i>Bg</i> ATP4 6F             | GGTAAAGAACAGATTTGCTGACATAGATCT   |
| <i>B<sup>g</sup></i> ATP4 6R  | ACCCTATTTCTTGTGCGATGGCCGTAGCTG   |
| <i>Bg</i> ATP4 7F             | AGGGACGCTATAAATACTTGTGGAAAGGC    |
| <i>B<sup>g</sup></i> ATP4 7R  | GTACGCCCAGACTCGATTGAGTTAACTATTG  |
| <i>Bg</i> ATP4 8F             | GGGTATTAATGGCACAGATGTTGCAAAAGG   |
| <i>B<sup>g</sup></i> ATP4 8R  | GACAGTTCCCACTAGCATCATTAAGTGTA    |
| <i>Bg</i> ATP4 9F             | ATCATCATTGCGCTATATGTTTCTACCGG    |
| <i>B<sup>g</sup></i> ATP4 9R  | CTAATGGGGTAGCATTTCCATTCTCAACAA   |
| <i>Bg</i> ATP4 10F            | AGGGAGTTCCTCCCATGTATAAAGATACCT   |
| <i>B<sup>g</sup></i> ATP4 10R | TTAGGCTTCTGCCCTTTTCATTTTTCGCTT   |

**Table supplement 1.**

**Primer sets of *B. gibsoni* ATP4**

| Receptor                       | Name    | Distance | Category               | Types                                    |
|--------------------------------|---------|----------|------------------------|------------------------------------------|
| <i>BgATP4</i> <sup>WT</sup>    | GLY407  | 2.67648  | Hydrogen Bond; Halogen | Carbon Hydrogen Bond; Halogen (Fluorine) |
|                                | CYS959  | 3.2396   | Halogen                | Halogen (Cl, Br, I)                      |
|                                | VAL962  | 2.66882  | Hydrophobic            | Pi-Sigma                                 |
|                                | CYS959  | 4.38742  | Hydrophobic            | Alkyl                                    |
|                                | VAL962  | 4.29249  | Hydrophobic            | Alkyl                                    |
|                                | ALA963  | 3.8231   | Hydrophobic            | Alkyl                                    |
|                                | ALA1202 | 3.67015  | Hydrophobic            | Alkyl                                    |
|                                | MET211  | 5.06565  | Hydrophobic            | Alkyl                                    |
|                                | VAL962  | 4.25668  | Hydrophobic            | Alkyl                                    |
|                                | ILE414  | 4.70605  | Hydrophobic            | Alkyl                                    |
|                                | CYS959  | 4.83891  | Hydrophobic            | Pi-Alkyl                                 |
|                                | VAL962  | 4.98402  | Hydrophobic            | Pi-Alkyl                                 |
|                                | VAL411  | 5.43239  | Hydrophobic            | Pi-Alkyl                                 |
|                                | VAL962  | 5.05537  | Hydrophobic            | Pi-Alkyl                                 |
|                                | PHE954  | 4.07586  | Hydrophobic            | Pi-Alkyl                                 |
| <i>BgATP4</i> <sup>L921V</sup> | GLY407  | 2.69013  | Hydrogen Bond; Halogen | Carbon Hydrogen Bond; Halogen (Fluorine) |
|                                | CYS959  | 3.22066  | Halogen                | Halogen (Cl, Br, I)                      |
|                                | VAL962  | 2.66391  | Hydrophobic            | Pi-Sigma                                 |
|                                | CYS959  | 4.35422  | Hydrophobic            | Alkyl                                    |
|                                | VAL962  | 4.31879  | Hydrophobic            | Alkyl                                    |
|                                | ALA963  | 3.8209   | Hydrophobic            | Alkyl                                    |
|                                | ALA1202 | 3.63038  | Hydrophobic            | Alkyl                                    |
|                                | MET211  | 5.0878   | Hydrophobic            | Alkyl                                    |
|                                | VAL962  | 4.24186  | Hydrophobic            | Alkyl                                    |
|                                | ILE414  | 4.71819  | Hydrophobic            | Alkyl                                    |
|                                | CYS959  | 4.82036  | Hydrophobic            | Pi-Alkyl                                 |
|                                | VAL962  | 4.99879  | Hydrophobic            | Pi-Alkyl                                 |
|                                | VAL411  | 5.4334   | Hydrophobic            | Pi-Alkyl                                 |
|                                | VAL962  | 5.04532  | Hydrophobic            | Pi-Alkyl                                 |
|                                | PHE954  | 4.07482  | Hydrophobic            | Pi-Alkyl                                 |
| <i>BgATP4</i> <sup>L921I</sup> | GLY958  | 2.92346  | Hydrogen Bond; Halogen | Carbon Hydrogen Bond; Halogen (Fluorine) |
|                                | GLY958  | 3.3301   | Halogen                | Halogen (Fluorine)                       |
|                                | MET410  | 5.88505  | Other                  | Pi-Sulfur                                |
|                                | VAL955  | 4.31406  | Hydrophobic            | Alkyl                                    |

**Table supplement 2.**

### Interactions of CIP from docking simulations



|         |         |             |          |
|---------|---------|-------------|----------|
| MET410  | 3.69023 | Hydrophobic | Alkyl    |
| VAL411  | 4.83101 | Hydrophobic | Alkyl    |
| VAL955  | 4.79513 | Hydrophobic | Alkyl    |
| VAL955  | 4.84729 | Hydrophobic | Alkyl    |
| CYS959  | 4.1269  | Hydrophobic | Alkyl    |
| ILE1205 | 4.82529 | Hydrophobic | Alkyl    |
| CYS959  | 5.21399 | Hydrophobic | Alkyl    |
| VAL955  | 4.90527 | Hydrophobic | Pi-Alkyl |
| PHE954  | 5.03012 | Hydrophobic | Pi-Alkyl |

**Table supplement 2.** (continued)

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## Editors

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### Reviewer #1 (Public review):

In this manuscript, the authors address an important issue in Babesia research by repurposing Cipargamin (CIP) as a potential therapeutic against selective Babesia spp. In this study, CIP demonstrated potent in vitro inhibition of B. bovis and B. gibsoni with IC50 values of 20.2 {plus minus} 1.4 nM and 69.4 {plus minus} 2.2 nM, respectively, and the in vivo efficacy against Babesia spp using mouse model. The authors identified two key resistance mutations in the BgATP4 gene (BgATP4L921I and BgATP4L921V) and explored their implications through phenotypic characterization of the parasite using cell biological experiments, complemented by in silico analysis. Overall, the findings are promising and could significantly advance Babesia treatment strategies.

#### Strengths:

In this manuscript, the authors effectively repurpose Cipargamin (CIP) as a potential treatment for Babesia spp. They provide compelling in vitro and in vivo data showing strong efficacy. Key resistance mutations in the BgATP4 gene are identified and analyzed through both phenotypic and in silico methods, offering valuable insights for advancing treatment strategies.

#### Weaknesses:

The manuscript explores important aspects of drug repurposing and rational drug design using Cipargamin (CIP) against Babesia. However, several weaknesses should be addressed. The study lacks novelty as similar research on Cipargamin has been conducted, and the experimental design could be improved. The rationale for choosing CIP over other ATP4-targeting compounds is not well-explained. Validation of mutations relies heavily on in silico predictions without sufficient experimental support. The Ion Transport Assay has limitations and would benefit from additional assays like Radiolabeled Ion Flux and Electrophysiological Assays. Also, the study lacks appropriate control drugs and detailed functional characterization. Further clarity on mutation percentages, additional safety testing, and exploration of cross-resistance would strengthen the findings.

(1) It is commendable to explore drug repurposing, drug deprescribing, drug repositioning, and rational drug design, especially using established ATP4 inhibitors that are well-studied in Plasmodium and other protozoan parasites. While the study provides some interesting findings, it appears to lack novelty, as similar investigations of Cipargamin on other

protozoan parasites have been conducted. The study does not introduce new concepts, and the experimental design could benefit from refinement to strengthen the results. Additionally, the rationale for choosing CIP over other MMV compounds targeting ATP4 is not clearly articulated. Clarifying the specific advantages CIP may offer against *Babesia* would be beneficial. Finally, the validation of the identified mutations might be strengthened by additional experimental support, as reliance on *in silico* predictions alone may not fully address the functional impact, particularly given the potential ambiguity of the mutations (BgATP4 L to V and I).

(2) Conducting an Ion Transport Assay is useful but has limitations. Non-specific binding or transport by other cellular components can lead to inaccurate results, causing false positives or negatives and making data interpretation difficult. Indirect measurements, like changes in fluorescence or electrical potential, can introduce artifacts. To improve accuracy, consider additional assays such as

- a. Radiolabeled Ion Flux Assay: tracks the movement of  $\text{Na}^+$  using radiolabeled ions, providing direct evidence of ion transport.
- b. Electrophysiological Assay: measures ionic currents in real-time with patch-clamp techniques, offering detailed information about ATP4 activity.

(3) *In-silico* predictions can provide plausible outcomes, but it is essential to evaluate how the recombinant purified protein and ligand interact and function at physiological levels. This aspect is currently missing and should be included. For example, incorporating immunoprecipitation and ATPase activity assays with both wild-type and mutant proteins, as well as detailed kinetic studies with Cipargamin, would be recommended to validate the findings of the study.

(4) The study lacks specific suitable control drugs tested both *in vitro* and *in vivo*. For accurate drug assessment, especially when evaluating drugs based on a specific phenotype, such as enlarged parasites, it is important to use ATP4 gene-specific inhibitors. Including similar classes of drugs, such as Aminopyrazoles, Dihydroisoquinolines, Pyrazoleamides, Pantothenamides, Imidazolopiperazines (e.g., GNF179), and Bicyclic Azetidine Compounds, would provide more comprehensive validation.

(5) Functional characterization of CIP through microscopic examination and quantification for assessing parasite size enlargement is not entirely reliable. A Flow Cytometry-Based Assay is recommended instead (along with suitable control antiparasitic drugs). To effectively monitor Cipargamin's action, conducting time-course experiments with 6-hour intervals is advisable rather than relying solely on endpoint measurements. Additionally, for accurate assessment of parasite morphology, obtaining representative qualitative images using Scanning Electron Microscopy (SEM) or Transmission Electron Microscopy (TEM) for treated versus untreated samples is recommended for precise measurements.

(6) A notable contradiction observed is that mutant cells displayed reduced efficacy and affinity but more pronounced phenotypic effects. The BgATP4L921I mutation shows a 2x lower susceptibility ( $\text{IC}_{50}$  of  $887.9 \pm 61.97$  nM) and a predicted binding affinity of  $-6.26$  kcal/mol with CIP. However, the phenotype exhibits significantly lower  $\text{Na}^+$  concentration in BgATP4L921I ( $P = 0.0087$ ) (Figure 3E).

(7) The manuscript does not clarify the percentage of mutations, and the number of sequence iterations performed on the ATP4 gene. It is also unclear whether clonal selection was carried out on the resistant population. If mutations are not present in 100% of the resistant parasites, please indicate the ratio of wild-type to mutant parasites and represent this information in the figure, along with the chromatograms.

(8) While the compound's toxicity data is well-established, it is advisable to include additional testing in epithelial cells and liver-specific cell lines (e.g., HeLa, HCT, HepG2) if feasible for the

authors. This would provide a more comprehensive assessment of the compound's safety profile.

(9) In the in vivo efficacy study, recrudescence parasites emerged after 8 days of treatment. Did these parasites harbor the same mutation in the ATP4 gene? The authors did not investigate this aspect, which is crucial for understanding the basis of recrudescence.

(10) The authors should explain their choice of Balb/c mice for evaluating CIP efficacy, as these mice clear the infection and may not fully represent the compound's effectiveness. Investigating CIP efficacy in SCID mice would be valuable, as they provide a more reliable model and eliminate the influence of the immune system. The rationale for not using SCID mice should be clarified.

(11) Do the in vitro-resistant parasites show any potential for cross-resistance with commonly used antiparasitic drugs? Have the authors considered this possibility, and what are their expectations regarding cross-resistance?

<https://doi.org/10.7554/eLife.101128.1.sa2>

#### **Reviewer #2 (Public review):**

##### Summary:

In this manuscript, the authors have tried to repurpose cipargamin (CIP), a known drug against plasmodium and toxoplasma against babesia. They proved the efficacy of CIP on babesia in the nanomolar range. In silico analyses revealed the drug resistance mechanism through a single amino acid mutation at amino acid position 921 on the ATP4 gene of babesia. Overall, the conclusions drawn by the authors are well justified by their data. I believe this study opens up a novel therapeutic strategy against babesiosis.

##### Strengths:

The authors have carried out a comprehensive study. All the experiments performed were carried out methodically and logically.

##### Weaknesses:

The introduction section needs to be more informative. The authors are investigating the binding of CIP to the ATP4 gene, but they did not give any information about the gene or how the ATP4 inhibitors work in general.

The resolution of the figures is not good and the font size is too small to read properly.

I also have several minor concerns which have been addressed in the "Recommendations for the authors" section.

<https://doi.org/10.7554/eLife.101128.1.sa1>

#### **Reviewer #3 (Public review):**

##### Summary:

The authors aim to establish that cipargamin can be used for the treatment of infection caused by Babesia organisms.

##### Strengths:

The study provides strong evidence that cipargamin is effective against various *Babesia* species. In vitro, growth assays were used to establish that cipargamin is effective against *Babesia bovis* and *Babesia gibsoni*. Infection of mice with *Babesia microti* demonstrated that cipargamin is as effective as the combination of atovaquone plus azithromycin. Cipargamin protected mice from lethal infection with *Babesia rodhaini*. Mutations that confer resistance to cipargamin were identified in the gene encoding ATP4, a P-type Na ATPase that was found in other apicomplexan parasites, thereby validating ATP4 as the target of cipargamin.

#### Weaknesses:

Cipargamin was tested in vivo at a single dose administered daily for 7 days. Despite the prospect of using cipargamin for the treatment of human babesiosis, there was no attempt to identify the lowest dose of cipargamin that protects mice from *Babesia microti* infection. Exposure to cipargamin can induce resistance, indicating that cipargamin should not be used alone but in combination with other drugs. There was no attempt at testing cipargamin in combination with other drugs, particularly atovaquone, in the mouse model of *Babesia microti* infection. Given the difficulty in treating immunocompromised patients infected with *Babesia microti*, it would have been informative to test cipargamin in a mouse model of severe immunosuppression (SCID or rag-deficient mice).

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